

FINAL PROJECT REPORT

Visualization Tool for Electric Vehicle Charge and Range Analysis

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Abstract

This project presents an interactive Tableau dashboard for analyzing electric vehicle charge, range, efficiency, pricing, and performance using a cleaned EV dataset.

Introduction

Electric vehicles are becoming increasingly popular. This dashboard helps users compare EV models through interactive visualizations.

Objectives

- Analyze battery range
- Compare charging speed
- Study pricing
- Compare manufacturers
- Evaluate efficiency and performance

Dataset Description

Source: Kaggle Electric Vehicle Dataset (CSV). Contains brand, model, battery, range, top speed, efficiency, charging speed, price and body style.

Software & Tools

Tableau Desktop, Tableau Public, Microsoft Excel.

Methodology

Data cleaning, preprocessing, worksheet creation, dashboard design, filters, publishing.

Dashboard Components

Top Brands by Range, Price vs Range, Rapid Charge Distribution, Body Style Distribution, Average Top Speed, Top Speed vs Price, Vehicle Segments, Efficiency vs Range, Price Distribution by Segment, Average Performance Score, Average Price per Km, Average Fast Charging Speed.

Key Findings

Tesla and Lucid offer high range; fast charging varies widely; SUVs dominate the dataset; price generally increases with performance.

Challenges

Missing values, inconsistent formatting, balancing dashboard readability.

Conclusion

The dashboard enables effective EV comparison and supports informed purchasing and research decisions.

Future Scope

Integrate live charging station data, predictive analytics, battery health monitoring, and real-time updates.

References

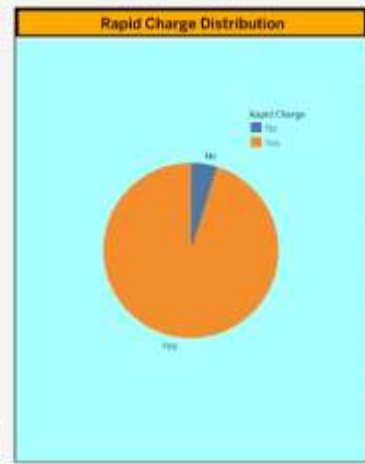
Kaggle EV Dataset

Tableau Public

Tableau Documentation

Dashboard Screenshot

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Brand

- Audi
- BMW
- Cadillac
- Chrysler
- DS
- Ford
- Genesis
- Hyundai
- Kia
- Lucid
- Mercedes
- Mini
- Nissan
- Porsche
- Subaru
- Tesla
- Volvo

